WASP-1b

R-1482

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_131023` · created 2026-08-01T13:01:13+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456588.7019 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.013
Transit depth (Rp/Rs)^2	1.07 +/- 0.27 [%]
Orbital Inclination (inc)	89.87 +/- 1.02
Airmass coefficient 1 (a1)	1.045 +/- 0.013
Airmass coefficient 2 (a2)	-0.041 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#3 - [288, 196]
Optimal Aperture	3.73
Optimal Annulus	10.18
Transit Duration (day)	0.1538 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.104 ± 0.013 vs 0.1036 (deviation 0.02σ)
Duration fit vs published	0.1538 d vs 0.1567 d (deviation 0.69σ)
Mid-time O–C	5.6 min
Depth SNR	4.0
Residual scatter	1.24 %

Light curve

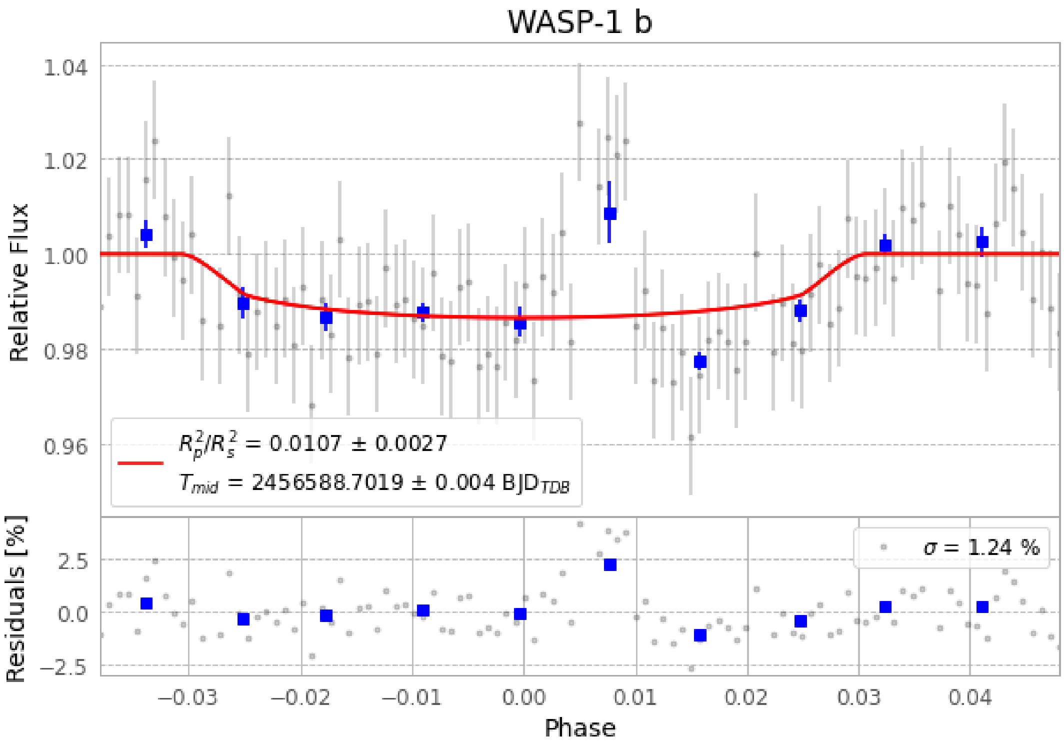

Figure 1 — FinalLightCurve_WASP-1 b_2013-10-22.png (SHA-256 b6e30bcf2543cfdf…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 235], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 aeac956115ffcba1…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

105 FITS frames (69 MB), `WASP-1131023022412.FITS` → `WASP-1131023073612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-131023.log (b6718c2fe1df…), FinalLightCurve_WASP-1 b_2013-10-22.png (b6e30bcf2543…), FinalLightCurve_WASP-1 b_2013-10-22.csv (05d6540fb716…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)